

Gaussian random vectors

- Independence
- Marginals and conditionals
- Demonstrations
- Credible ellipses

Normal distributions

- A **multivariate normal** random variable is denoted with

$$w \sim \text{Normal}_D(\mu, V)$$

The probability density function is given by

$$p(w) = \text{Normal}_D(w; \mu, V) = \frac{1}{\sqrt{(2\pi)^D |V|}} \exp\left(-\frac{(w - \mu)^t V^{-1} (w - \mu)}{2}\right)$$

- A **univariate normal** random variables is denoted with

$$w \sim \text{Normal}(\mu, v)$$

The probability density function is given by

$$p(w) = \text{Normal}(w; \mu, v) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{1}{2} \frac{(w - \mu)^2}{v}\right)$$

Independence

We consider a **special multivariate** random variable

$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \sim \text{Normal}_D \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{pmatrix}, \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_D \end{bmatrix} \right), \quad w_d \in \mathbb{R}, \quad \mu_d \in \mathbb{R}, \quad v_d \in \mathbb{R}_+$$

The probability density is

$$p \left(\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \right) = \prod_{d=1}^D \text{Normal}(w_d; \mu_d, v_d) \implies \begin{cases} w_d \sim \text{Normal}(\mu_d, v_d) \\ d = 1, \dots, D \end{cases}$$

Similarly, we also show that

$$\left\{ w_d \sim \text{Normal}(\mu_d, v_d) \right. \\ \left. d = 1, \dots, D \right\} \implies \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \sim \text{Normal}_D \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{pmatrix}, \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_D \end{bmatrix} \right)$$

Independence

We consider a **special multivariate** random variable

$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \sim \text{Normal}_D \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{pmatrix}, \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_D \end{bmatrix} \right), \quad w_d \in \mathbb{R}, \quad \mu_d \in \mathbb{R}, \quad v_d \in \mathbb{R}_+$$

According to the probability densities, we have the **equivalence**

$$\left. \begin{matrix} w_d \sim \text{Normal}(\mu_d, v_d) \\ d = 1, \dots, D \end{matrix} \right\} \iff \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \sim \text{Normal}_D \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{pmatrix}, \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_D \end{bmatrix} \right)$$

- One consequence is that

$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \sim \text{Normal}_D(0, I) \iff \begin{cases} w_d \sim \text{Normal}(0, 1) \\ d = 1, \dots, D \end{cases}$$

- By convention, we will be writing

$$p \left(\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} \right) \equiv p(w_1, w_2, \dots, w_D)$$

Marginals

We consider a **multivariate** random variable and a linear transformation

$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \sim \text{Normal}_3 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \right), \quad \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

Because of the **general transformation property**, we have

$$\begin{aligned} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} &\sim \text{Normal}_2 \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^t \right) \\ &= \text{Normal}_2 \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \right) \\ &= \text{Normal}_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \right) \end{aligned}$$

We construct similarly **any** other multivariate normal **marginal**

Rearrangements

We consider a **multivariate** random variable and a linear transformation

$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \sim \text{Normal}_3 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \right), \quad \begin{pmatrix} w_3 \\ w_1 \\ w_2 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

Because of the **general transformation property**, we have

$$\begin{aligned} \begin{pmatrix} w_3 \\ w_1 \\ w_2 \end{pmatrix} &\sim \text{Normal}_3 \left(\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^t \right) \\ &= \text{Normal}_3 \left(\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \right) \\ &= \text{Normal}_3 \left(\begin{pmatrix} \mu_3 \\ \mu_1 \\ \mu_2 \end{pmatrix}, \begin{bmatrix} v_{33} & v_{13} & v_{23} \\ v_{13} & v_{11} & v_{12} \\ v_{23} & v_{21} & v_{22} \end{bmatrix} \right) \end{aligned}$$

We construct similarly **any** other multivariate normal **rearrangement**

Conditionals

We consider a **multivariate** random variable partitioned in two **blocks**

$$\begin{pmatrix} W_1 \\ W_2 \end{pmatrix} \sim \text{Normal}_D \left(\begin{pmatrix} M_1 \\ M_2 \end{pmatrix}, \begin{bmatrix} V_1 & C^t \\ C & V_2 \end{bmatrix} \right), \quad D_1 + D_2 = D \quad \begin{matrix} W_1 \in \mathbb{R}^{D_1} & M_1 \in \mathbb{R}^{D_1} & V_1 \in \mathbb{R}^{D_1 \times D_1} \\ W_2 \in \mathbb{R}^{D_2} & M_2 \in \mathbb{R}^{D_2} & V_2 \in \mathbb{R}^{D_2 \times D_2} \end{matrix} \quad C \in \mathbb{R}^{D_2 \times D_1}$$

By the definition of conditionals, we have

$$p(W_1|W_2) = \frac{p(W_1, W_2)}{p(W_2)} = \frac{\text{Normal}_D \left(\begin{pmatrix} W_1 \\ W_2 \end{pmatrix}; \begin{pmatrix} M_1 \\ M_2 \end{pmatrix}, \begin{bmatrix} V_1 & C^t \\ C & V_2 \end{bmatrix} \right)}{\text{Normal}_{D_2}(W_2; M_2, V_2)} = \dots = \text{Normal}_{D_1}(W_1; M'_1, V'_1)$$

We obtain similarly **any** other multivariate normal **conditional**

$$W_1|W_2 \sim \text{Normal}_{D_1}(M'_1, V'_1)$$

The conditional parameters are

$$\begin{aligned} M'_1 &= M_1 + C^t V_2^{-1} (W_2 - M_2) \\ V'_1 &= V_1 - C^t V_2^{-1} C \end{aligned}$$

Bayesian considerations

We consider a **Bayesian model** with scalar observations and obtain its posterior **sequentially**

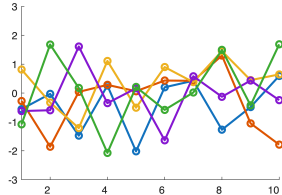
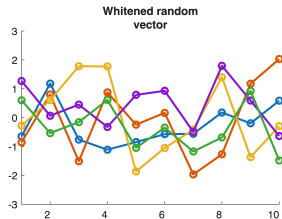
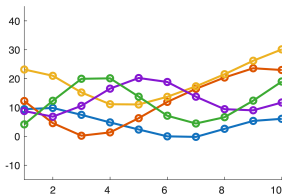
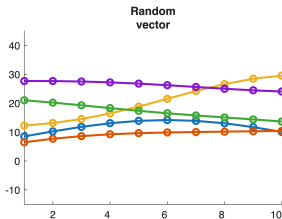
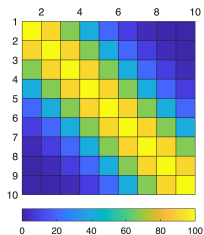
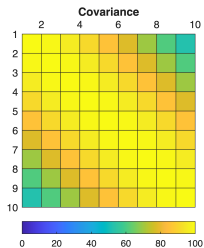
$$\left. \begin{array}{l} \mu \sim \text{Normal}_D(M, V) \\ w_1 | \mu \sim \text{Normal}(\beta_1 \mu, v_1) \\ w_2 | \mu \sim \text{Normal}(\beta_2 \mu, v_2) \end{array} \right\} \Rightarrow \left. \begin{array}{l} \mu | w_1 \sim \text{Normal}_D(M', V') \\ w_2 | \mu \sim \text{Normal}(\beta_2 \mu, v_2) \end{array} \right\} \Rightarrow \mu | w_1, w_2 \sim \text{Normal}_D(M'', V'')$$

Equivalently

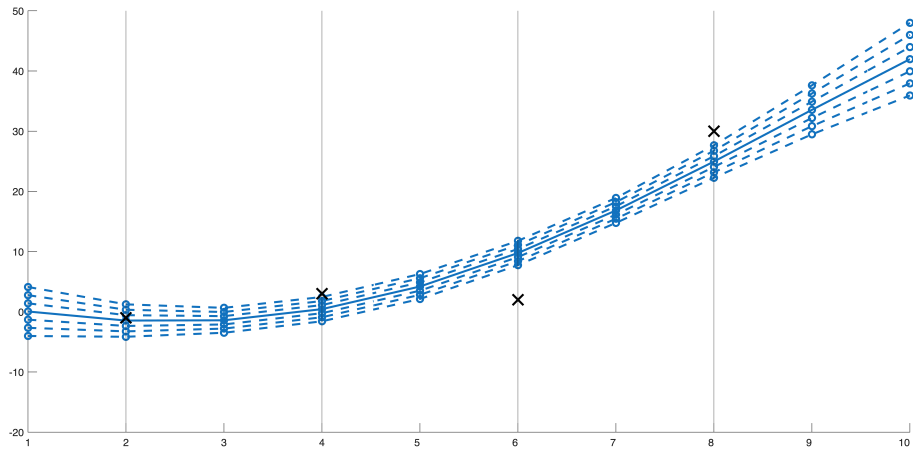
We consider a **Bayesian model** with vector observations and obtain its posterior **directly**

$$\left. \begin{array}{l} \mu \sim \text{Normal}_D(M, V) \\ \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} | \mu \sim \text{Normal}_2 \left(\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \mu, \begin{bmatrix} v_1 & 0 \\ 0 & v_2 \end{bmatrix} \right) \end{array} \right\} \Rightarrow \mu | \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \sim \text{Normal}_D(M'', V'')$$

Whitening transformations



Sequential updates



Bivariate normal distributions

- A **bivariate normal** random variable is denoted with

$$w \sim \text{Normal}_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{bmatrix} v_1 & \rho\sqrt{v_1v_2} \\ \rho\sqrt{v_1v_2} & v_2 \end{bmatrix} \right), \quad w_1, w_2 \in \mathbb{R}, \quad \mu_1, \mu_2 \in \mathbb{R}, \quad v_1, v_2 \in \mathbb{R}_+, \quad \rho \in (-1, 1)$$

A **credible set** is a set of values with a designated total probability of containing the value of our uncertain quantity

The γ -credible ellipse is an **ellipse** with a total probability $\gamma \in [0, 1]$

This is given by

$$w_1(t) = \mu_1 + \sqrt{sv_1} \cos(t)$$

$$w_2(t) = \mu_2 + \sqrt{sv_2} \left(\rho \cos(t) + \sqrt{1 - \rho^2} \sin(t) \right)$$

$$t \in [0, 2\pi)$$

for $s = -2 \log(1 - \gamma)$

